Cmod A7-35T Internal IP Peripheral Reference

Platform: Xilinx Artix-7 (xc7a35tcpg236-1) — Cmod A7-35T
Processor: MicroBlaze RISC-V
System Clock: 100 MHz (PLL from 12 MHz on-board oscillator via Clocking Wizard)
Toolchain: Vivado & Vitis 2025.2

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1. Processor Core & System Infrastructure

1.1 MicroBlaze RISC-V (`microblaze_riscv_0`)

Parameter	Value
IP Version	`xilinx.com:ip:microblaze_riscv:1.0`
Instruction / Data Bus	32-bit LMB (Local Memory Bus) + AXI4 Data Port
Debug Interface	Enabled (`C_DEBUG_ENABLED = 1`)
Multiply / Divide	Enabled (`C_USE_MULDIV = 2`)
Bit Manipulation	Bitmanip A/B/C/S all enabled
Cache	16 KB I-Cache + 16 KB D-Cache, write-through, 32 B lines, cacheable range `0x6000_0000` – `0x6007_FFFF` (exact SRAM fit)

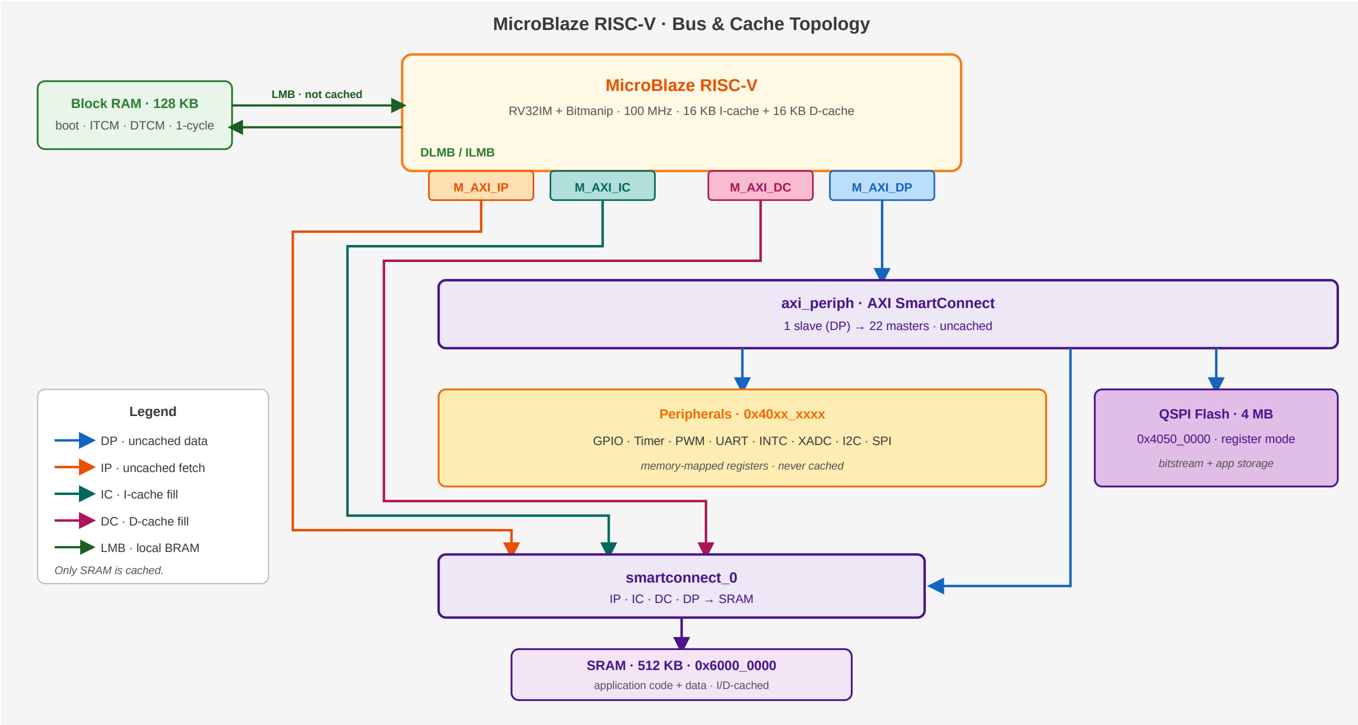
Description: The main processor core. Executes user firmware and accesses all peripherals through the AXI SmartConnect interconnect.

1.2 Local Memory

Parameter	Value
IP	`lmb_v10` (LMB Bus) + `lmb_bram_if_cntlr` + `blk_mem_gen`
Data Bus (DLMB)	0x0000_0000 – 0x0001_FFFF (128 KB)
Instruction Bus (ILMB)	0x0000_0000 – 0x0001_FFFF (128 KB)
Memory Type	True Dual-Port Block RAM
ECC	Disabled

Description: Instruction and data local memory implemented with FPGA Block RAM — functionally this MCU's tightly-coupled memory (TCM): the ILMB/DLMB ports play the same role as ITCM/DTCM on other MCU cores (e.g. Cortex-M7), giving 1-cycle access outside the cache path. Layout: `0x0000` – `0x7FFF` (32 KB) holds the UART bootloader (restored from flash at every configuration); `0x8000` – `0xFFFF` (32 KB) is the application's ITCM for interrupt handlers and timing-critical code (`ITCM_FUNC`, copied out of the SRAM image by `tcm_init()` at startup); `0x10000` – `0x1FFFF` (64 KB) is the DTCM, holding the stack (top-down from `0x20000`) and `DTCM_DATA` fast data.

1.3 AXI SmartConnect (`microblaze_riscv_0_axi_periph`)



Parameter	Value
IP Version	`xilinx.com:ip:smartconnect:1.0`
Master Ports	22 (M00 – M21)
Slave Ports	1 (S00, connected to MicroBlaze M_AXI_DP)

Description: AXI interconnect crossbar that routes processor data transactions to 22 peripheral endpoints.

1.4 AXI Interrupt Controller (`microblaze_riscv_0_axi_intc`)

Parameter	Value
IP Version	`xilinx.com:ip:axi_intc:4.1`
AXI Base Address	`0x4040_0000`
Fast Interrupt	Disabled (`C_HAS_FAST = 0`)
Interrupt Sources	8 (merged via `ilconcat`)

Interrupt Mapping:

Channel	Source	Description

Channel	Source	Description
In0	`timer_0`	System timer 0 interrupt
In1	`timer_1`	System timer 1 interrupt
In2	`timer_2`	System timer 2 interrupt
In3	`uart_1`	External UART interrupt
In4	`uart_USB`	USB UART interrupt
In5	`INT_0_3`	External GPIO interrupt (4-bit)
In6	`i2c_0`	I2C controller interrupt
In7	`spi_0`	External SPI master interrupt

1.5 Debug Module (`mdm_1`)

Parameter	Value
IP Version	`xilinx.com:ip:mdm_riscv:1.0`

Description: MicroBlaze Debug Module providing JTAG debug access for breakpoints, register inspection, and memory read/write. Its `Debug_SYS_Rst` signal is connected to the system reset module.

1.6 Clocking Wizard (`clk_wiz_1`)

Parameter	Value
IP Version	`xilinx.com:ip:clk_wiz:6.0`
Input Clock	12 MHz (on-board oscillator)
Output Clock	100 MHz
PLL Multiply Factor	MMCM_CLKFBOUT_MULT_F = 62.5
PLL Divide Factor	MMCM_CLKOUT0_DIVIDE_F = 7.5

Description: Uses an MMCM/PLL to multiply the 12 MHz board clock to 100 MHz for the entire system. The `locked` signal indicates clock stability.

1.7 Processor System Reset (`rst_clk_wiz_1_100M`)

Parameter	Value
IP Version	`xilinx.com:ip:proc_sys_reset:5.0`
External Reset	On-board push button (Active Low)

Description: Generates synchronized reset signals (`mb_reset`, `bus_struct_reset`, `peripheral_aresetn`) ensuring all modules are released from reset only after the clock is stable.

2. UART Communication

2.1 USB UART (`uart_USB`)

Parameter	Value
IP Version	`xilinx.com:ip:axi_uart16550:2.0`
AXI Base Address	`0x4030_0000`
Address Range	64 KB (0x4030_0000 – 0x4030_FFFF)
TX Pin	J18 (via Micro-USB connector)
RX Pin	J17 (via Micro-USB connector)
Interrupt	Connected to `xlconcat In4`

Description: 16550-compatible UART for host PC communication through the on-board Micro-USB connector. Baud rate is software-configured via the Divisor Latch Register. At 100 MHz system clock, the divisor for 115200 baud is 54 (0x36). Typically mapped as STDIN/STDOUT.

2.2 External UART (`uart_1`)

Parameter	Value
IP Version	`xilinx.com:ip:axi_uart16550:2.0`
AXI Base Address	`0x4031_0000`
Address Range	64 KB (0x4031_0000 – 0x4031_FFFF)
TX Pin	J1 (DIP Pin 11)
RX Pin	K2 (DIP Pin 12)
Interrupt	Connected to `xlconcat In3`

Description: Second 16550 UART exposed on the DIP connector for communication with external devices (e.g., sensor modules, Bluetooth modules).

3. GPIO (General Purpose I/O)

All GPIO instances use `xilinx.com:ip:axi_gpio:2.0`.

3.1 On-Board GPIO

Instance	AXI Base Address	Width	Direction	Connection	Description
`board_led_2bits`	`0x4000_0000`	2	Output	A17, C16	On-board LEDs × 2

Instance	AXI Base Address	Width	Direction	Connection	Description
`board_button`	`0x4001_0000`	1	Input	A18	On-board push button × 1
`board_rgb`	`0x4002_0000`	3	Output	B17, B16, C17	On-board RGB LED (R/G/B)

3.2 DIP Connector GPIO (4 Groups × 7-bit)

Instance	AXI Base Address	Width	DIP Pins	Description
`gpio_A_0_6`	`0x4003_0000`	7	Pin 1–7	GPIO Group A, bidirectional I/O
`gpio_B_0_6`	`0x4004_0000`	7	Pin 17–23	GPIO Group B, bidirectional I/O
`gpio_C_0_6`	`0x4005_0000`	7	Pin 42–48	GPIO Group C, bidirectional I/O
`gpio_D_0_6`	`0x4006_0000`	7	Pin 26–32	GPIO Group D, bidirectional I/O

Description: Each group is a 7-bit bidirectional port (`C_ALL_OUTPUTS = 0`). Pin direction is set through the TRI register; data is read/written via the DATA register.

3.3 External Interrupt Inputs (`INT_0_3`)

Parameter	Value
AXI Base Address	`0x4007_0000`
Width	4-bit (all inputs)
Interrupt Capability	Enabled (`C_INTERRUPT_PRESENT = 1`)
DIP Pins	Pin 8 (INTR_0), Pin 9 (INTR_1), Pin 41 (INTR_2), Pin 33 (INTR_3)

Description: Four external interrupt inputs grouped into a single AXI GPIO instance. The interrupt signal is routed to `xlconcat In5` and then to the AXI Interrupt Controller.

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4. Timers & PWM

All timer instances use `xilinx.com:ip:axi_timer:2.0`.

4.1 System Timers

Instance	AXI Base Address	Mode	Interrupt	Description
`timer_0`	`0x4010_0000`	32-bit (`Default`)	`xlconcat In0`	General-purpose system timer

Instance	AXI Base Address	Mode	Interrupt	Description
`timer_1`	`0x4011_0000`	Default	`xlconcat In1`	General-purpose timer
`timer_2`	`0x4012_0000`	Default	`xlconcat In2`	General-purpose timer

4.2 PWM Outputs

Instance	AXI Base Address	Output Pin	DIP Pin	Description
`PWM_0`	`0x4020_0000`	J3	Pin 10	PWM Channel 0
`PWM_1`	`0x4021_0000`	W3	Pin 34	PWM Channel 1
`PWM_2`	`0x4022_0000`	W4	Pin 40	PWM Channel 2

Description: AXI Timer instances configured in PWM mode to generate square-wave outputs. Useful for LED dimming, motor speed control, buzzer tone generation, etc. Frequency and duty cycle are configured through the Timer Load Registers and the PWM enable bit.

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5. External Memory

5.1 QSPI Flash (`axi_quad_spi_0`)

Parameter	Value
IP Version	`xilinx.com:ip:axi_quad_spi:3.2`
AXI_LITE Base (Control)	`0x4050_0000` (64 KB)
Mode	Standard SPI controller (`C_XIP_MODE = 0`) — **not** memory-mapped
SPI Memory Type	Macronix (`C_SPI_MEMORY = 4`)
Interface	Quad SPI, FIFO depth 256

Description: Controls the on-board Quad-SPI NOR Flash. The controller runs in **standard (register) mode**: the full register set (control, status, TX/RX FIFO, slave select — see PG153) is exposed at `0x4050_0000` via `microblaze_riscv_0_axi_periph` M19, so the CPU can issue any SPI command — read (`0x0B`), Write Enable (`0x06`), Sector/Block Erase (`0x20` / `0xD8`), Page Program (`0x02`), status poll (`0x05`). This is what allows the UART bootloader to program application images into flash at runtime (`workspace-example/bootloader/src/bootloader.c` is a complete worked example, and the Vitis `XSpi` driver wraps the register protocol).

Flash contents are **not memory-mapped** in this mode — there is no XIP window, and code cannot execute from flash directly. The standalone-boot design instead copies the application from flash into SRAM at power-on (see the Standalone Boot Mode guide). The 256-entry FIFO matches the flash's 256-byte page size, so one page program fits in a single transfer.

> **Design note:** XIP mode (`C_XIP_MODE = 1`, a read-only memory-mapped window) and CPU-programmable register mode are **mutually exclusive** in this IP (PG153). This project chooses register mode: self-programming (Arduino-style `upload.py` workflow) was judged more valuable for the course than execute-in-place, and the 512 KB SRAM + I-cache serves execution instead.

5.2 SRAM / Cellular RAM (`axi_emc_0`)

Parameter	Value
IP Version	`xilinx.com:ip:axi_emc:3.0`
AXI Base Address	`0x6000_0000`
Address Range	512 KB, exact fit (0x6000_0000 – 0x6007_FFFF)
Physical Capacity	512 KB Cellular RAM

Description: External memory controller providing access to the on-board 512 KB SRAM. Suitable for large data buffers, but access latency is higher than Block RAM.

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6. Analog-to-Digital Conversion (XADC)

6.1 XADC Wizard (`xadc_wiz_0`)

Parameter	Value
IP Version	`xilinx.com:ip:xadc_wiz:3.3`
AXI Base Address	`0x4060_0000`
Address Range	64 KB
Conversion Rate	500 KSPS
Sequencer Mode	Continuous

Enabled Channels:

Channel	Source	Description
VAUX4	DIP Pin 15 (G3/G2)	External analog input 0 (on-board divider: 0–3.3 V → 0–1 V)
VAUX12	DIP Pin 16 (H2/J2)	External analog input 1 (on-board divider: 0–3.3 V → 0–1 V)
Temperature	Internal	FPGA die temperature monitor
VCCINT	Internal	Core voltage monitor (1.0 V)
VCCAUX	Internal	Auxiliary voltage monitor (1.8 V)

Description: The Artix-7 built-in 12-bit ADC capable of measuring external analog signals and monitoring internal FPGA temperature and supply voltages. External input pins pass through an on-board resistive voltage divider (2.32 KΩ / 1 KΩ, ratio ≈ 0.301), accepting up to 3.3 V at the DIP pin.

Effective Per-Channel Sampling Rate: The sequencer continuously cycles through all 5 enabled channels (VAUX4, VAUX12, Temperature, VCCINT, VCCAUX). The aggregate conversion rate is 500 KSPS, giving each channel an effective rate of 500 K ÷ 5 = **100 KSPS**.

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7. Serial Expansion Interfaces (I2C / SPI)

7.1 I2C Controller (`i2c_0`)

Parameter	Value
IP Version	`xilinx.com:ip:axi_iic:2.1`
AXI Base Address	`0x4070_0000`
SCL Frequency	100 kHz (standard mode)
Input Glitch Filter	50 ns on SCL and SDA (`C_*_INERTIAL_DELAY = 5` @ 100 MHz) — per the I2C tSP spike-suppression spec
SCL / SDA Pins	DIP Pin 13 (L1) / Pin 14 (L2)
Pull-ups	Weak FPGA internal pull-ups enabled in XDC; **external 4.7 kΩ to 3.3 V recommended** for real devices
Interrupt	Connected to `xlconcat In6`

Description: AXI IIC master for external I2C devices (sensors, EEPROMs, OLED displays). Open-drain signaling: any device may only pull the line low; the pull-up resistor returns it high. Devices are addressed by their 7-bit I2C address. Use the Vitis `XIic` driver.

7.2 External SPI Master (`spi_0`)

Parameter	Value
IP Version	`xilinx.com:ip:axi_quad_spi:3.2` (standard mode, no STARTUP)
AXI Base Address	`0x4080_0000`
SCLK	100 MHz / 16 = **6.25 MHz** (fixed at synthesis, `C_SCK_RATIO = 16`)
Slave Selects	2 (`C_NUM_SS_BITS = 2`) — two devices can share the bus
Pins	DIP 35 SCLK (V3) · 36 MOSI (W5) · 37 MISO (V4) · 38 SS0 (U4) · 39 SS1 (V5)
Interrupt	Connected to `xlconcat In7`

Description: SPI master for external devices (displays, ADCs, flash modules). Full-duplex push-pull signaling — no pull-ups needed; the active device is chosen by driving its SS line low. Use the Vitis `XSpi` driver.

> **Note:** This is a *second, independent* SPI controller — do not confuse it with `axi_quad_spi_0` (`0x4050_0000`), which is dedicated to the on-board QSPI boot flash. Firmware should select controllers by instance macro (`XPAR_SPI_0_BASEADDR` vs `XPAR_AXI_QUAD_SPI_0_BASEADDR`), never by generic `XPAR_XSPI_n_*` numbering.

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8. Complete Address Map

Peripheral addresses follow a class-based convention — **`0x40[C]x_xxxx`, where `C` is the peripheral class** (1 MB per class, 64 KB per instance). Reading an address immediately tells you what kind of device it is: class 0 = GPIO, 1 = Timer, 2 = PWM, 3 = UART, 4 = INTC, 5 = QSPI control, 6 = XADC, 7 = I2C, 8 = SPI.

AXI Base Address	Range	Peripheral	IP Type	Category
`0x0000_0000`	128K / 128K	Local Memory (BRAM)	blk_mem_gen	Memory
`0x4000_0000`	64K	board_led_2bits	axi_gpio	GPIO
`0x4001_0000`	64K	board_button	axi_gpio	GPIO
`0x4002_0000`	64K	board_rgb	axi_gpio	GPIO
`0x4003_0000`	64K	gpio_A_0_6	axi_gpio	GPIO
`0x4004_0000`	64K	gpio_B_0_6	axi_gpio	GPIO
`0x4005_0000`	64K	gpio_C_0_6	axi_gpio	GPIO
`0x4006_0000`	64K	gpio_D_0_6	axi_gpio	GPIO
`0x4007_0000`	64K	INT_0_3	axi_gpio	Interrupt
`0x4010_0000`	64K	timer_0	axi_timer	Timer
`0x4011_0000`	64K	timer_1	axi_timer	Timer
`0x4012_0000`	64K	timer_2	axi_timer	Timer
`0x4020_0000`	64K	PWM_0	axi_timer	PWM
`0x4021_0000`	64K	PWM_1	axi_timer	PWM
`0x4022_0000`	64K	PWM_2	axi_timer	PWM
`0x4030_0000`	64K	uart_USB	axi_uart16550	Communication
`0x4031_0000`	64K	uart_1	axi_uart16550	Communication
`0x4040_0000`	64K	axi_intc	axi_intc	System

AXI Base Address	Range	Peripheral	IP Type	Category
`0x4050_0000`	64K	axi_quad_spi_0 (`AXI_LITE`, full register set)	axi_quad_spi	Memory
`0x4060_0000`	64K	xadc_wiz_0	xadc_wiz	ADC
`0x4070_0000`	64K	i2c_0 (DIP 13/14)	axi_iic	Communication
`0x4080_0000`	64K	spi_0 (external master, DIP 35–39)	axi_quad_spi	Communication
`0x6000_0000`	512K	axi_emc_0 (exact physical fit, I/D-cached)	axi_emc	Memory